Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.04

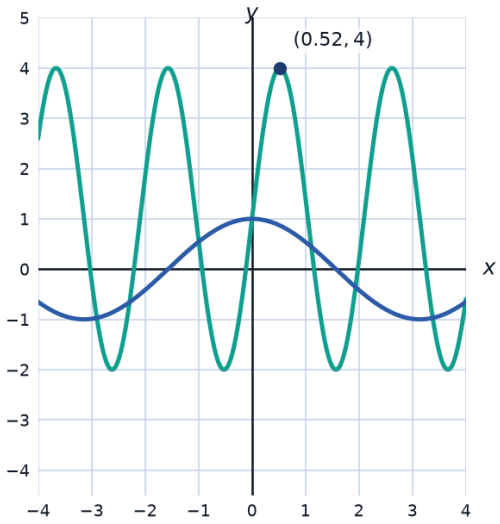
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

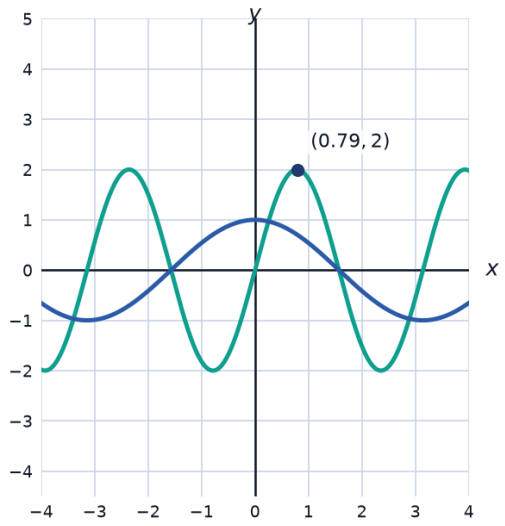
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

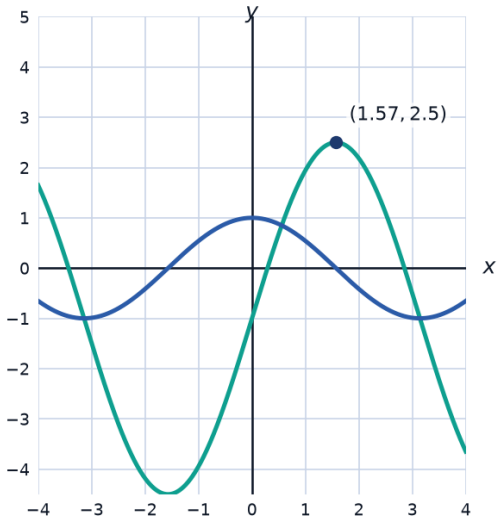
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Indices, Surds & Quadratics · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Differentiation & Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Simultaneous Equations & Inequalities · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

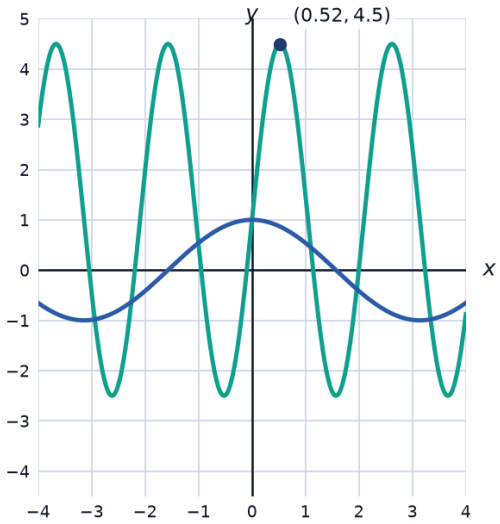
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Further Integration Techniques

[Trigonometry · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.02 Differential Equations**

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Matrices (AI HL)**

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.03 Markov Chains (AI HL)**

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the domain and range of the principal arcsine function $y = \arcsin x$?

a. Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

b. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

c. Domain: $[-1, 1]$; Range: $[0, \pi]$

d. Domain: $(-\infty, \infty)$; Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the domain and range of $y = \arccos x$?

a. Domain: $[-1, 1]$; Range: $[0, \pi]$

b. Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

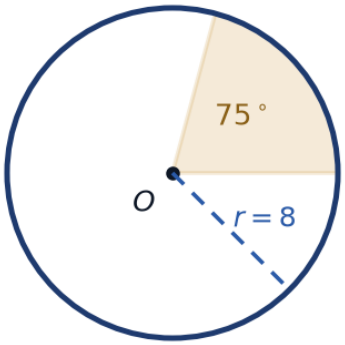
c. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

d. Domain: $[0, 1]$; Range: $[0, \pi/2]$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the horizontal asymptotes of the graph of $y = \arctan x$?



a. $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$

b. $y = \pi$ and $y = 0$

c. $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

d. $y = 1$ and $y = -1$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

a. $\frac{3}{\sqrt{1-9x^2}}$

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{\sqrt{1-3x^2}}$

d. $\frac{3}{1+9x^2}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

a. $\frac{2}{4+x^2}$

b. $\frac{1}{1+x^2/4}$

c. $\frac{1}{4+x^2}$

d. $\frac{2}{\sqrt{4-x^2}}$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

a. $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$

b. $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

c. $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

d. They are reciprocals of each other

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

a. 1

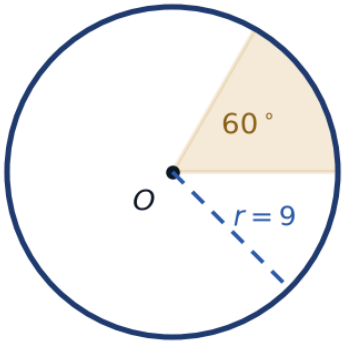
b. 0

c. $\pi/2$

d. Undefined

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?



a. The slope approaches $+\infty$ (vertical tangent)

b. The slope approaches 0

c. The slope approaches 1

d. The slope approaches -1

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

a. $y - \frac{\pi}{4} = \frac{1}{2}(x - 1)$

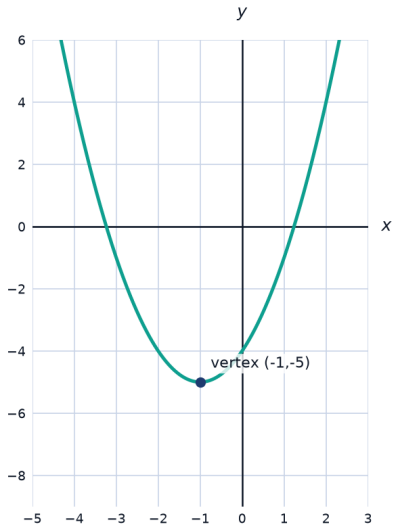
b. $y - \frac{\pi}{4} = 1(x - 1)$

c. $y = \frac{1}{2}x + \frac{\pi}{4}$

d. $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 5 units down

b. Shifting 1 units right and 5 units down

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

For what value of k is $f(x) = -2x + k$ (for $x < 5$) and $f(x) = x^2$ (for $x \geq 5$) continuous at $x = 5$?

a. 36

b. 34

c. -35

d. 35

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

If $f(x) = \sqrt{x - 5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \leq x \leq 3$

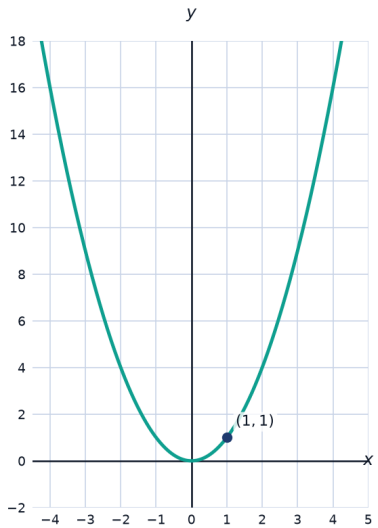
b. $x \leq -3$ or $x \geq 3$

c. $x \geq 3$

d. All real numbers

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.



a. -6

b. -4

c. 9

d. -5

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. Odd

c. Neither

d. Cannot be determined

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ or $5 < k < 11$

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$

b. $k = -2;\ x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

a. $x = 0$ and $x = 4$

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Matrices (AI HL)

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$

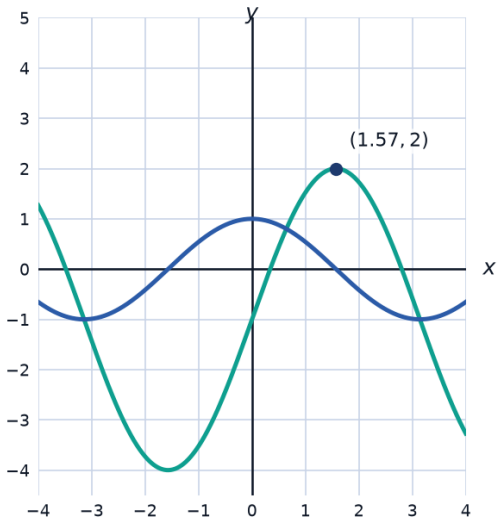
b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?



a. Graph A

b. Graph B

c. Graph C

d. Graph D

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.03 Markov Chains (AI HL)

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1-x^2}$

b. $\frac{1}{\sqrt{1-x^2}}$

c. $1-x^2$

d. $\sqrt{x^2-1}$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Proof by Mathematical Induction

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$

d. $-\frac{3}{2}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

If $\sin\theta = -\frac{20}{29}$ and θ is in Quadrant 3, find $\cos\theta$.

a. $\frac{29}{21}$

b. $\frac{21}{29}$

c. $-\frac{21}{29}$

d. $\frac{20}{29}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.04 Partial Fractions

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. Lagrange multiplier

b. eigenvalue

c. gradient

d. curvature

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

If $f'(x) = (x - 3)$, on which interval is f increasing?

a. $(3, \infty)$

b. \{3\}

c. $(-\infty, \infty)$

d. $(-\infty, 3)$

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